EXPERIMENT - 3

AIM: Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

SOURCE CODE:

```
CREATE TABLE Department ( dept_id
   INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

CREATE TABLE Project (
   project_id INT PRIMARY KEY,
   project_name VARCHAR(50),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Employee ( emp_id
   INT PRIMARY KEY,
   emp_name VARCHAR(50),
   dept_id INT, project_id
   INT, salary
   DECIMAL(10,2), city
   VARCHAR(50),
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
   FOREIGN KEY (project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
(1, 'HR'), (2, 'Finance'), (3, 'IT'), (4, 'Marketing'), (5, 'Operations');

INSERT INTO Project VALUES
(101, 'Recruitment System', 1), (102, 'Payroll Automation', 2),
(103, 'Cybersecurity Upgrade', 3), (104, 'Website Revamp', 3),
(105, 'Ad Campaign', 4), (106, 'Market Research', 4),
(107, 'Logistics Optimization', 5), (108, 'Inventory System', 5);

INSERT INTO Employee VALUES
(1, 'Alice', 1, 101, 45000, 'Delhi'),
(2, 'Bob', 1, 101, 42000, 'Mumbai'),
(3, 'Charlie', 1, 101, 47000, 'Pune'),
(4, 'David', 2, 102, 55000, 'Delhi'),
(5, 'Eva', 2, 102, 60000, 'Lucknow'),
(6, 'Frank', 2, 102, 52000, 'Chennai'),
(7, 'Grace', 3, 103, 70000, 'Bangalore'),
(8, 'Hannah', 3, 103, 68000, 'Hyderabad'),
(9, 'Ian', 3, 104, 72000, 'Delhi'),
(10, 'Jack', 3, 104, 65000, 'Pune'),
(11, 'Karen', 3, 104, 64000, 'Mumbai'),
(12, 'Leo', 4, 105, 48000, 'Delhi'),
```

```
(13, 'Mia', 4, 105, 50000, 'Lucknow'),
(14, 'Nina', 4, 106, 51000, 'Chennai'),
(15, 'Oscar', 4, 106, 53000, 'Delhi'),
(16, 'Paul', 5, 107, 56000, 'Hyderabad'),
(17, 'Quinn', 5, 107, 57000, 'Bangalore'),
(18, 'Rita', 5, 107, 59000, 'Delhi'),
(19, 'Sam', 5, 108, 60000, 'Lucknow'),
(20, 'Tina', 5, 108, 61000, 'Mumbai'),
(21, 'Uma', 5, 108, 62000, 'Chennai'),
(22, 'Victor', 2, 102, 53000, 'Delhi'),
(23, 'Wendy', 2, 102, 54000, 'Hyderabad'),
(24, 'Xavier', 3, 103, 71000, 'Bangalore'),
(25, 'Yara', 3, 104, 66000, 'Lucknow'),
(26, 'Zane', 4, 105, 49000, 'Delhi'),
(27, 'Aditi', 4, 106, 52000, 'Mumbai'),
(28, 'Bhavya', 1, 101, 46000, 'Chennai'),
(29, 'Chetan', 1, 101, 43000, 'Lucknow'),
(30, 'Devika', 5, 107, 58000, 'Delhi');

SELECT *
FROM Employee
WHERE city = 'Delhi';

SELECT emp_name, salary
FROM Employee;

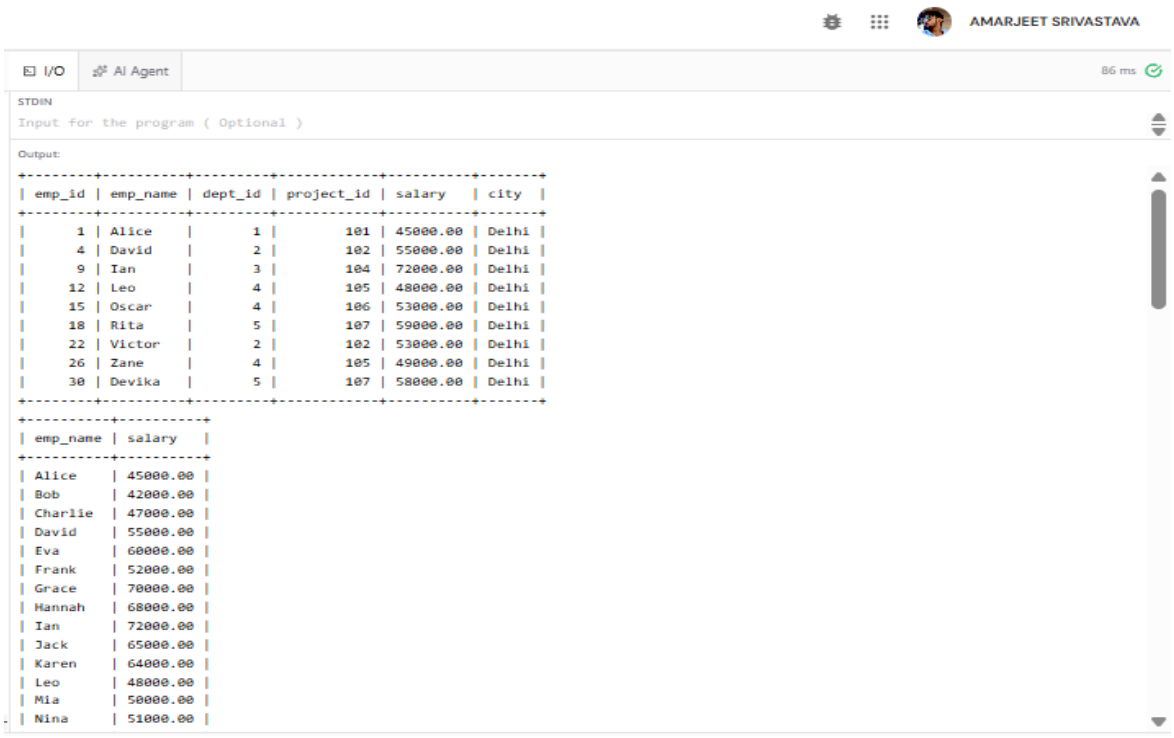
SELECT dept_id, AVG(salary) AS avg_salary
FROM Employee
GROUP BY dept_id;

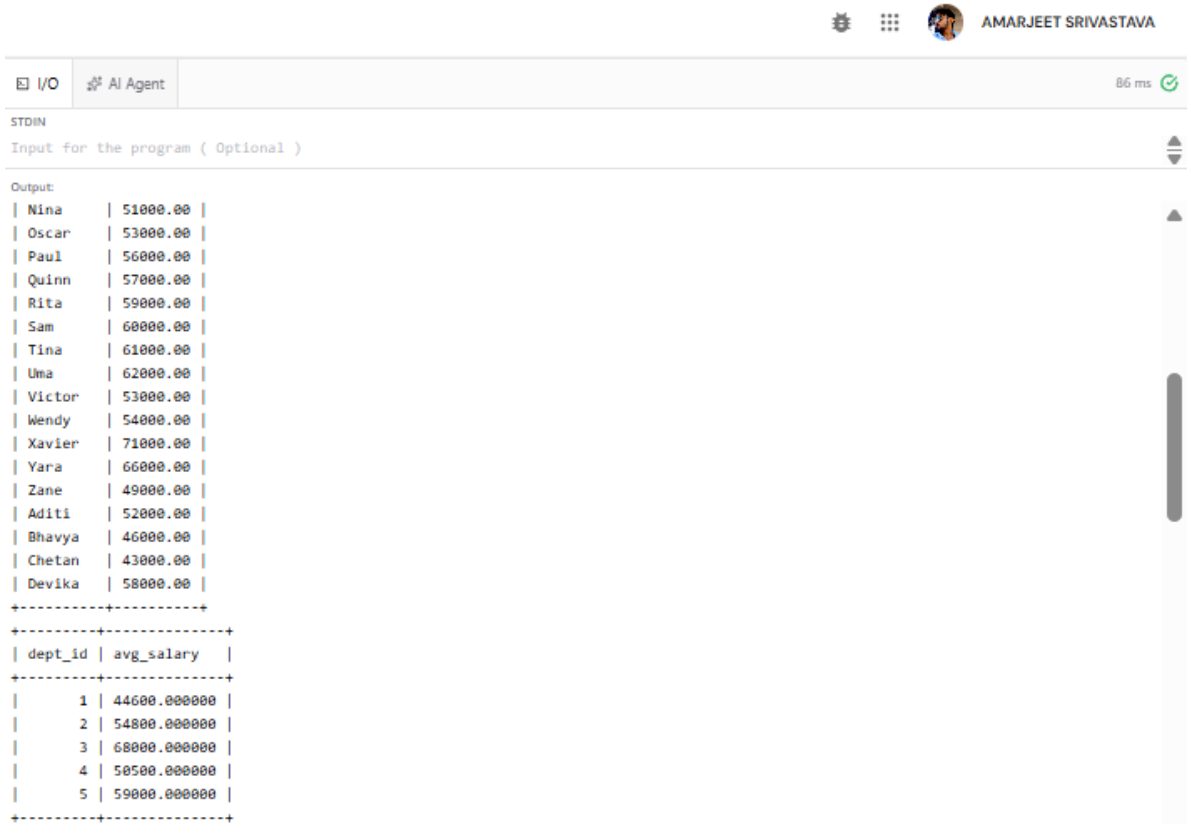
SELECT dept_id, COUNT(*) AS num_employees
FROM Employee
GROUP BY dept_id
HAVING COUNT(*) > 5;

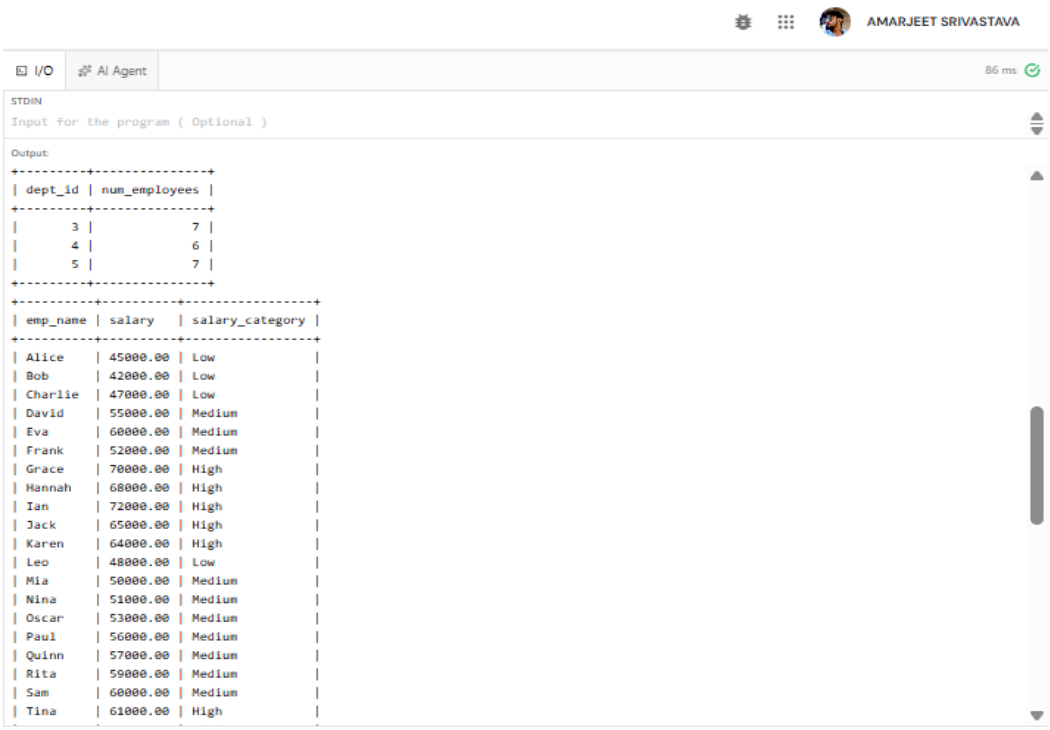
SELECT emp_name, salary,
    CASE
        WHEN salary < 50000 THEN 'Low'
        WHEN salary BETWEEN 50000 AND 60000 THEN 'Medium'
        ELSE 'High'
    END AS salary_category
FROM Employee;

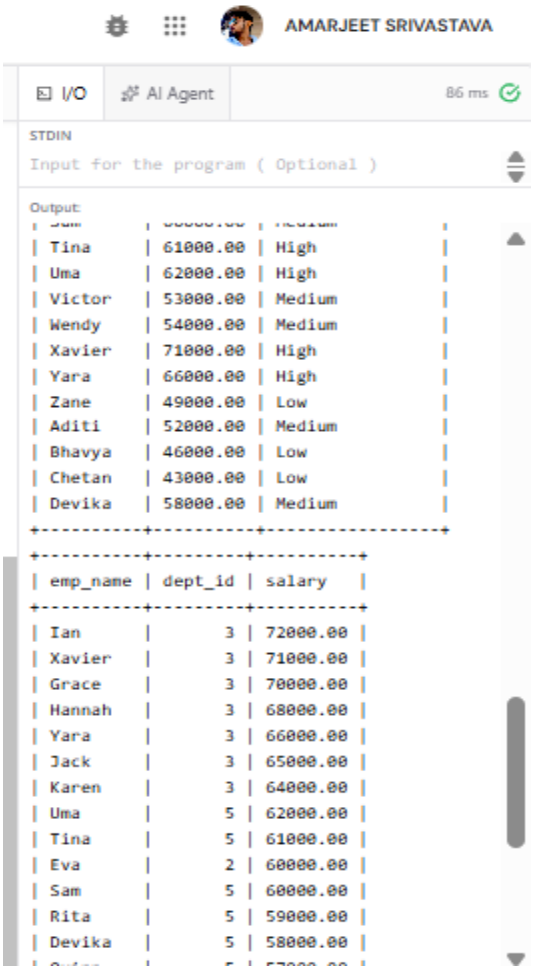
SELECT emp_name, dept_id, salary
FROM Employee
ORDER BY salary DESC;
```

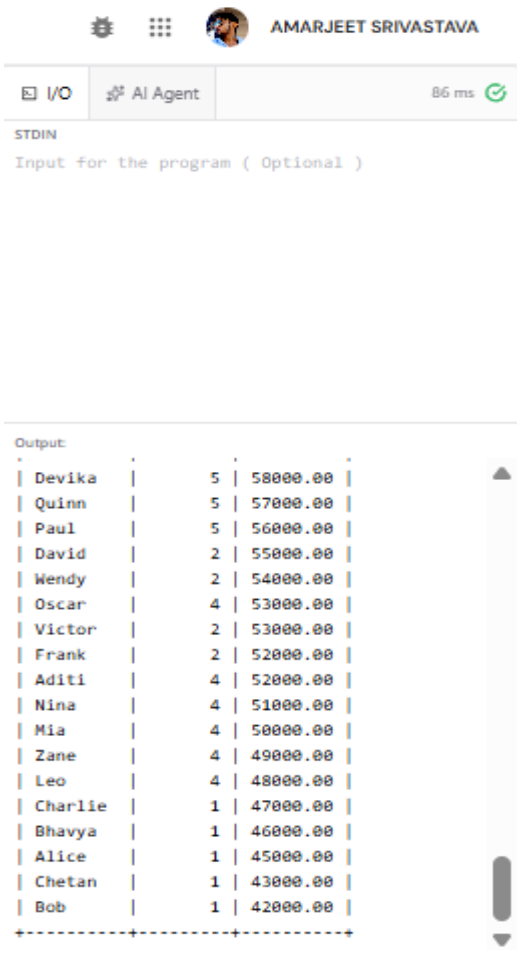
OUTPUT:


AMARJEET SRIVASTAVA
I/O
AI Agent
86 ms
STDIN
Input for the program (Optional)
Output:
+--------+----------+---------+------------+----------+-------+
| emp_id | emp_name | dept_id | project_id | salary | city |
+--------+----------+---------+------------+----------+-------+
1	Alice	1	101	45000.00	Delhi
4	David	2	102	55000.00	Delhi
9	Ian	3	104	72000.00	Delhi
12	Leo	4	105	48000.00	Delhi
15	Oscar	4	106	53000.00	Delhi
18	Rita	5	107	59000.00	Delhi
22	Victor	2	102	53000.00	Delhi
26	Zane	4	105	49000.00	Delhi
30	Devika	5	107	58000.00	Delhi
+--------+----------+---------+------------+----------+-------+
+----------+----------+
| emp_name | salary |
+----------+----------+
Alice	45000.00
Bob	42000.00
Charlie	47000.00
David	55000.00
Eva	60000.00
Frank	52000.00
Grace	70000.00
Hannah	68000.00
Ian	72000.00
Jack	65000.00
Karen	64000.00
Leo	48000.00
Mia	50000.00
Nina	51000.00


AMARJEET SRIVASTAVA
I/O
AI Agent
86 ms
STDIN
Input for the program (Optional)
Output:
Nina	51000.00
Oscar	53000.00
Paul	56000.00
Quinn	57000.00
Rita	59000.00
Sam	60000.00
Tina	61000.00
Uma	62000.00
Victor	53000.00
Wendy	54000.00
Xavier	71000.00
Yara	66000.00
Zane	49000.00
Aditi	52000.00
Bhavya	46000.00
Chetan	43000.00
Devika	58000.00
+----------+----------+
+---------+--------------+
| dept_id | avg_salary |
+---------+--------------+
1	44600.000000
2	54800.000000
3	68000.000000
4	50500.000000
5	59000.000000
+---------+--------------+


AMARJEET SRIVASTAVA
I/O
AI Agent
86 ms
STDIN
Input for the program (Optional)
Output:
+---------+---------------+
| dept_id | num_employees |
+---------+---------------+
3	7
4	6
5	7
+---------+---------------+
+----------+---------+-----------------+
| emp_name | salary | salary_category |
+----------+---------+-----------------+
Alice	45000.00	Low
Bob	42000.00	Low
Charlie	47000.00	Low
David	55000.00	Medium
Eva	60000.00	Medium
Frank	52000.00	Medium
Grace	70000.00	High
Hannah	68000.00	High
Ian	72000.00	High
Jack	65000.00	High
Karen	64000.00	High
Leo	48000.00	Low
Mia	50000.00	Medium
Nina	51000.00	Medium
Oscar	53000.00	Medium
Paul	56000.00	Medium
Quinn	57000.00	Medium
Rita	59000.00	Medium
Sam	60000.00	Medium
Tina	61000.00	High


AMARJEET SRIVASTAVA
I/O
AI Agent
86 ms
STDIN
Input for the program (Optional)
Output:
Tina	61000.00	High
Uma	62000.00	High
Victor	53000.00	Medium
Wendy	54000.00	Medium
Xavier	71000.00	High
Yara	66000.00	High
Zane	49000.00	Low
Aditi	52000.00	Medium
Bhavya	46000.00	Low
Chetan	43000.00	Low
Devika	58000.00	Medium
+----------+----------+------------------+
+----------+---------+----------+
| emp_name | dept_id | salary |
+----------+---------+----------+
Ian	3	72000.00
Xavier	3	71000.00
Grace	3	70000.00
Hannah	3	68000.00
Yara	3	66000.00
Jack	3	65000.00
Karen	3	64000.00
Uma	5	62000.00
Tina	5	61000.00
Eva	2	60000.00
Sam	5	60000.00
Rita	5	59000.00
Devika	5	58000.00


AMARJEET SRIVASTAVA
I/O
AI Agent
86 ms
STDIN
Input for the program (Optional)
Output:
Devika	5	58000.00
Quinn	5	57000.00
Paul	5	56000.00
David	2	55000.00
Wendy	2	54000.00
Oscar	4	53000.00
Victor	2	53000.00
Frank	2	52000.00
Aditi	4	52000.00
Nina	4	51000.00
Mia	4	50000.00
Zane	4	49000.00
Leo	4	48000.00
Charlie	1	47000.00
Bhavya	1	46000.00
Alice	1	45000.00
Chetan	1	43000.00
Bob	1	42000.00
+----------+---------+----------+